One obstruction, three applications: supply chains, ontologies, and agent orchestration as composition problems for directed containers

A grant-impact outlook (staged, not for the book)

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Abstract

A directed container is exactly a small category; this is machine-checked in our Lean development. It follows that anything which presents a small category — a supply-chain network, an ontology log, an orchestration of agents — *is* a directed container, and composing two such systems is one and the same mathematical problem: the existence of a Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$. We have proved that this product exists if and only if a pairwise criterion $(L) \wedge (G)$ holds, and that — under a mild hypothesis on the shared node — its only obstruction is a single degree-two cohomology class $[\omega] \in H^2(\mathrm{Sk}_\mathcal{C}; \mathcal{D})$. This outlook argues that one theorem thereby governs three superficially unrelated failures — supply-chain inventory inconsistency, ontology-merge conflict, and agent re-entrancy — and that in the supply-chain instance the class refines from a single bit to a $\mathbb{Z}/n$-valued count of how badly two provenance routes disagree. We are honest about grades throughout: the general theorems are proved (the core is Lean-verified), the domain instantiations are computed worked examples, and whether a *real* supply chain literally presents a category remains an open modelling question. This is the Impact spine of the Kodamai grant, presented as a program, not a finished chapter.

1 Introduction

Three questions, from three different rooms of the building.

Logistics. Two supply chains share a warehouse. Can their flows be serialized into one consistent schedule, or do the two routes to a delivered good disagree about where it came from?

Knowledge engineering. Two teams maintain two ontologies that share a concept. Can the schemas be merged, or do they clash on a convention?

Agent systems. A meta-agent orchestrates two sub-agents that both touch a shared resource. Does the orchestration serialize, or does a re-entrant call corrupt the token?

These look like three engineering problems for three specialists. This note argues they are *one* mathematical problem with *one* obstruction, and that we have already proved the theorem that resolves it. The bridge is a single machine-checked fact.

The hinge. A *directed container* in the sense of Ahman–Uustalu [1] is exactly a small category: shapes are objects, positions out of a shape are the morphisms out of that object, the root is the identity, and the shift operation is composition. We have formalised this equivalence in Lean 4 (`DContCat.lean`, sorry-free) [LEAN-VERIFIED]. Its consequence is a licence: *anything that presents* *a small category is a directed container, for free*, and every structural result we have about directed containers — which functors are containers, the four monoidal products, the comonoid classifications — applies to it immediately.

The claim of this note. Supply chains, ontology logs, and orchestrated agents all present categories (with one honest caveat, below). Composing two of them — wiring two chains, merging two ontologies, sequencing two agents — is therefore composing two directed containers, which is the problem of forming a *Zappa–Szép product* $\mathcal{C} \bowtie \mathcal{D}$ (an internal-to-**Cat** two-sided semidirect product; equivalently a distributive law $\mathcal{DC} \Rightarrow \mathcal{CD}$, equivalently a strict factorization system). That a distributive law of directed containers assembles this product — generalizing the monoid Zappa–Szép product — is Ahman–Uustalu's [2]; what we add is a characterization of *when* it exists. We have proved:

(P1) **Existence criterion** [3] [PROVED]. The product $\mathcal{C} \bowtie \mathcal{D}$ exists iff a pairwise condition holds: (L) every hom-set is a free right $\mathcal{D}$-set, and (G) the free bases close into a wide subcategory.

(P2) **Obstruction theorem** [4] [PROVED]. Under (L) and a mild hypothesis (abelian vertex groups, trivial left action), (G) holds iff a cohomology class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ vanishes, where $\mathrm{Sk}_{\mathcal{C}}$ is the orbit category and $\mathcal{D}$ the vertex-group presheaf.

So the composite serializes iff $[\omega] = 0$, and $[\omega] \neq 0$ *is* the failure: the inventory inconsistency, the merge conflict, the re-entrancy bug. One class, three faces. That is the sentence this note exists to make believable, and it is the Impact spine of the grant: the applications are not new mathematics but new *instances of one proved theorem*.

What is ours and what is borrowed. The equivalence DCont $\simeq$ SmallCat is Ahman–Uustalu [1] (we contribute the Lean proof). The Zappa–Szép-product construction itself — a distributive law of directed containers assembling the joint container, generalizing the monoid case — is likewise Ahman–Uustalu's [2]. Ontology logs are Spivak's prior art [9]; the reading of a knowledge-graph instance as a Set-valued copresheaf on its schema is folklore categorical databases. What is *ours* is the *existence-and-obstruction* layer over that construction: the pairwise criterion (P1) saying when the product exists, the H^2 obstruction (P2), and their first proved instance — agent orchestration [5], whose core computation $[\omega] = \varepsilon$ is itself Lean-verified (`Reentrancy.lean`) [LEAN-VERIFIED]. The delta this note stakes is running that obstruction theory *through* logistics and ontologies, where our corpus currently has only analogy.

Grades up front. This note mixes results at very different levels of certainty, and the honest thing is to say which is which before the reader invests any trust.

> **Grade key.** [LEAN-VERIFIED] machine-checked in Lean 4. [PROVED] proved on paper (registry grade). [COMPUTED] verified on an explicit finite instance by computer. [DEFINITIONAL IMPORT] a definition imported from prior art, no new theorem. **[open]** an unresolved modelling question.
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> *Nothing about a domain rises above* [COMPUTED]*. The object-level claim "a real supply chain is a category" stays* **[open]** *(SEED Q4). We build faithful minimal abstractions and let the proved theorems compute on them; we never assert that reality is the abstraction.*

Outline. Section 2 states the hinge. Section 3 places the three domains inside the equivalence, grading each. Section 4 states the composition theorems and reports the computed supply-chain instance, including its $\mathbb{Z}/n$ refinement and the ontology sibling. Section 5 is the honest status table and the outlook.

2 The hinge: a directed container is a small category

Theorem 1 (Ahman–Uustalu [1]; our Lean proof [LEAN-VERIFIED])**.** *The category* **DCont** *of directed containers is equivalent to the category of small categories. Explicitly, a directed container* $(S, P, o, \downarrow, \oplus)$ *is the data of a small category: objects* S*, morphisms out of* s *the positions* $P(s)$*, identities the roots* o_s*, composition the shift* $\oplus$*, and the five directed-container laws are the category laws.*

We have machine-checked both round trips of this dictionary in `DContCat.lean` (sorry-free). The point for what follows is not the proof but the *licence* it grants:

> *To exhibit a system as a directed container, it suffices to present it as a small category.*

Every downstream structure comes along for free. In particular the morphisms of **DCont** are not functors but *cofunctors* (equivalently, the Put half of a delta lens [8]): an object map forward, and for each object a backward lift of outgoing moves. This is the correct variance for the applications — a map of systems propagates states forward and updates backward — and we return to it in Section 3.

3 Three domains present categories

3.1 Agent orchestration [proved]

A collection of agents with handoffs between them is a small category: agents (or agent states) are objects, permitted handoffs are morphisms, sequencing is composition. A meta-agent supervising sub-agents is a cofunctor: it routes request-shapes down and lifts a sub-agent's response-move back to a meta-level move [8]. This is the domain where we first ran the full machine: orchestration composition is a Zappa–Szép product, and its re-entrancy obstruction is the class $[\omega]$, proved and Lean-verified [5]. It is the parity anchor ($n = 2$) for the general picture below.

3.2 Supply chains [open] at the object level, [proved] at the morphism level

Neil's phrasing is that "supply chains present categories." Concretely: stages or locations (supplier, factory, warehouse, customer) are objects, operations or flows (procure, manufacture, ship) are morphisms, do-then-do is composition, and the consistency / inventory-conservation conditions are the path-equations (two routes delivering the same good must agree — the sheaf condition of the grant's Path 5).

The honest content is a *conditional*: a supply chain is a directed container *iff* it genuinely presents a category — operations compose associatively and the consistency path-equations hold on the nose. Whether a real network does so is a modelling question, SEED Open Question 4, and stays **[open]**. What is settled is the *morphism* level: a map between supply-chain models is a cofunctor / delta-lens Put, hence a directed- container morphism [PROVED] [7] — the get is forward state propagation, the put is consistent backward update propagation. So the maps are pinned down even while the objects await a fidelity theorem.

3.3 Ontology logs and knowledge graphs [definitional import]

An *olog* [9] is a finite category schema: types are objects ("a book," "an author"), aspects are morphisms ("a book *has* an author"), and facts are commuting-path equations. A knowledge-graph instance or database over the schema is a functor $\mathcal{C} \to \mathbf{Set}$: a Set-valued copresheaf sending each type to its set of entities and each aspect to a function. Both are prior art; we import the definitions and do not reinvent them.[1]

The bridge is then definitional: by Theorem 1 an olog, being a small category, *is* a directed container, with types as shapes, available aspects as positions, aspect composition as $\oplus$, and facts as the position-equations [DEFINITIONAL IMPORT]. No new theorem — but it is worth a clean statement because it places knowledge graphs *inside* the container equivalence chain, so the whole structure theory becomes available to them. A survey of our corpus on 23 July 2026 found the word "olog" entirely absent: this is greenfield for us.

4 The payoff: composition is a Zappa–Szép product, obstructed in H^2

Now the existence-and-obstruction layer that is ours. Merging or wiring two of these systems is composing two directed containers over a shared piece — a shared warehouse, a shared type, a shared resource. That is exactly the problem of forming a Zappa–Szép product — the directed-container construction of Ahman–Uustalu [2] — and the question we answer is *when* such a product exists and what obstructs it.

Theorem 2 (Existence, (P1) [3] [PROVED])**.** *Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory. There is a wide $\mathcal{C}$ with $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ — equivalently a distributive law $\mathcal{DC} \Rightarrow \mathcal{CD}$, equivalently a strict factorization system $(\mathcal{C}, \mathcal{D})$ — iff* (L) *every* $\mathrm{Hom}_{\mathcal{K}}(-, b)$ *is a free right $\mathcal{D}$-set and* (G) *the free bases close into a wide subcategory.*

Theorem 3 (Obstruction, (P2) [4] [PROVED])**.** *Under* (L) *and hypothesis* (H) *(abelian vertex groups, trivial left action), (G) $\Leftrightarrow [\omega] = 0 \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$, with $\mathrm{Sk}_{\mathcal{C}}$ the orbit category and $\mathcal{D}$ the vertex-group presheaf. The cohomology is that of Baues–Wirsching / Rosebrugh–Wood, cited as classical background, not reproved.*

Chaining the two: *the composite serializes iff* $[\omega] = 0$. When $[\omega] \neq 0$ no Zappa–Szép product exists — the systems cannot be consistently interleaved — and the nonzero class is the concrete failure in each domain. This is the whole story in one line; the rest of this section makes it concrete on a computed example.

4.1 A computed supply-chain instance [computed]

We have carried this out on the minimal honest instance [6]; the numerical claims below are machine-checked in `scratch/supply_chain_zs.py`.

Take the procurement chain $\mathsf{Src} \to \mathsf{Mfg} \to \mathsf{Dst}$ as an explicit poset category and write out its directed container: shapes are the three stages, positions out of a stage are its outgoing operations, the root is "stay put," and $\oplus$ is do-then-do (procure-then-ship *is* the direct-delivery operation). The five directed-container laws hold by an exhaustive check [COMPUTED].

[1] Provenance flag: our reading log has Spivak–Kent [9] at *agent-summary* grade only. We cite it by name for the *definition* of an olog — a definitional import, no theorem number is claimed — and mark a deep-read of the primary source as a TODO before this material is promoted anywhere beyond staging.

Now let two flows meet at a warehouse that keeps an internal *lot-cursor*: a cyclic pointer τ of order n into n interchangeable lot-slots. The cursor moves nothing physically — it is pure provenance bookkeeping — but a unit leaving carries whatever cursor value was current, and $\mathrm{End(Wh)} \cong \mathbb{Z}/n$. Two routes to the same delivered good, a nominal line and a rework line, may advance the cursor differently; a single datum $\varepsilon \in \mathbb{Z}/n$ records the discrepancy. Fixing the warehouse-internal relabelings as the right factor $\mathcal{D}$, hypotheses (L) and (H) hold uniformly in ε, the orbit category and presheaf are computed, and:

Theorem 4 (The class is the unit-mismatch [6] [COMPUTED])**.** *For this warehouse abstraction $\mathcal{W}_{n,\varepsilon}$,*

$$H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/n, \qquad [\omega(\mathcal{W}_{n,\varepsilon})] = \varepsilon.$$

Hence $\mathcal{W}_{n,\varepsilon} = \mathcal{C} \bowtie \mathcal{D}$ exists iff $\varepsilon = 0$, i.e. iff the two routes agree on lot-provenance. When $\varepsilon \neq 0$ the obstruction $[\omega] = \varepsilon$ is a nonzero element of $\mathbb{Z}/n$: a genuine inventory inconsistency, the two routes disagreeing by ε lot-positions, with no serialization possible. An independent brute-force cross-check counts the strict factorization systems: n of them when $\varepsilon = 0$, none when $\varepsilon \neq 0$.

Remark 5 (A bit becomes a counter)**.** Agent orchestration [5] needed only $\mathbb{Z}/2$: a turn token is one bit. Inventory is naturally a *counter*, and the obstruction lands in the counter's group. The class $[\omega] = \varepsilon$ is not merely "inconsistent yes/no" but *by how many units* the two provenance routes disagree, valued in $\mathbb{Z}/n$. The gauge freedom (the choice of which slot is "slot zero") shifts both routes together, so only their difference ε is observable — the categorical avatar of a flat $\mathbb{Z}/n$-bundle being trivial iff its holonomy vanishes. The proved parity theorem is the $n = 2$ face of this refinement.

4.2 The ontology sibling [computed]

The same machine at $n = 2$ is the merge of two ologs sharing a type. Merging the schema $\mathsf{Book} \to \mathsf{Author} \to \mathsf{Name}$ with a second olog sharing Author is composing two directed containers over the shared type. Let the shared type carry a $\mathbb{Z}/2$ token — a choice of naming convention ("Last, First" vs. "First Last"). Then Theorem 4 at $n = 2$ reads: the merge succeeds iff the two schemas agree on the convention; a disagreement is the nonzero class $[\omega] = 1$, a merge conflict — the *same* H^2 bit as supply-chain inconsistency and agent re-entrancy [COMPUTED].

5 Status, and what it buys the grant

Honest status table.

Claim	Grade	Where
DCont $\simeq$ SmallCat (the hinge)	[LEAN-VERIFIED]	`DContCat.lean` [1]
composition = ZS; exists $\iff (L) \wedge (G)$	[PROVED]	[3]
failure = $[\omega] \in H^2$ (the obstruction)	[PROVED]	[4]
orchestration re-entrancy = $[\omega] = \varepsilon$	[LEAN-VERIFIED]	`Reentrancy.lean` [5]
supply-chain map = cofunctor / lens	[PROVED]	[7]
computed supply-chain ZS instance ($\mathbb{Z}/n$)	[COMPUTED]	[6]
olog / KG-schema = small category = DCont	[DEFINITIONAL IMPORT]	[9]; this note
a real supply chain *presents* a category	**[open]**	SEED Q4

The grant line. Supply-chain composability, ontology-merge consistency, and agent-orchestration re-entrancy are the *same* degree-two obstruction on the handoff category. In the logistics instance the invariant refines from a bit to a $\mathbb{Z}/n$-valued count of the discrepancy. This is exactly the shape of Impact the grant needs: not three ad hoc categorical analogies but three instances of one proved theorem, with the core machine- checked, and with a clear frontier — the object-level modelling criterion — honestly marked open.

Open questions. Two are worth stating precisely.

- *The fidelity criterion (SEED Q4).* Give checkable conditions under which a real supply-chain network genuinely presents a category — associativity of operations and the inventory-consistency path-equations. This upgrades the standing analogy to a criterion and is the one [**open**] row in the table.
- *Beyond abelian vertex groups.* Theorem 3 assumes hypothesis (H). For richer bookkeeping monoids at a shared node (non-abelian relabelings, non-invertible state), what replaces $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$? The answer would extend the obstruction from these finite instances to the full nonabelian setting the applications will eventually demand.

Status of this note. Staging for the applications turn of the grant, produced for Neil and Robin. It commits nothing to the book. The general theorems it leans on are cited at their registry grades; the domain instantiations are computed; the fidelity claim is open. When Neil steers the applications turn, this is ready to slot in.

References

[1] D. Ahman, T. Uustalu, *Directed containers as categories*, EPTCS 207 (2016), 89–98. (Our Lean proof: `DContCat.lean`, sorry-free.)

[2] D. Ahman, T. Uustalu, *Distributive laws of directed containers*, Progress in Informatics, No. 10 (2013), 3–18, DOI 10.2201/NIIPI.2013.10.2. (Source of the Zappa–Szép-product construction: a distributive law of directed containers assembles the joint container, generalizing the monoid Zappa–Szép product.)

[3] MacBeth, *The pairwise Zappa–Szép criterion*, Kodamai note, 10 June 2026 (proved).

[4] MacBeth, *The global-closure obstruction* (G) *is a cohomology class*, Kodamai note, 11 June 2026 (proved).

[5] MacBeth, *The re-entrancy obstruction is the token-mutation bit: an analytic proof for orchestration Zappa–Szép products*, Kodamai note, 20 July 2026 (proved; $[\omega] = \varepsilon$ core Lean-verified, `Reentrancy.lean`).

[6] MacBeth, *A supply chain is a directed container, and composing two is a Zappa–Szép product: a computed worked example*, Kodamai note, 23 July 2026 (computed; `scratch/supply_chain_zs.py`).

[7] MacBeth, *Directed-container morphisms are cofunctors, i.e. the Put of a delta lens*, Kodamai connection note, 9–11 June 2026 (proved; building on Clarke [8]).

[8] B. Clarke, *Internal lenses as functors and cofunctors*, EPTCS 323 (2020), 183–195. (Cited by name for the cofunctor / delta-lens terminology.)

[9] D. I. Spivak, R. E. Kent, *Ologs: a categorical framework for knowledge representation*, PLoS ONE 7(1) (2012), e24274. *[Provenance: cited by name for the definition of an olog; deep-read of the primary source is a standing TODO before promotion beyond staging.]*